

Line Balancing, SAM Values and Capacity Policy

Document SOP-PR-04 | Type: production_procedure | Effective: 2026-02-10

Standard allowed minutes by style

Capacity is derived from the standard allowed minutes (SAM) of each style and the number of operators on the line. A line of 30 operators working an eight hour shift at 85 percent efficiency yields 12,240 productive minutes.

SKU	Style	SAM	Realistic daily output per 30 operators
GAR-001	Classic Black Polo	12.5	980 units
GAR-002	White Cotton T-Shirt	8.0	1,530 units
GAR-003	Navy Formal Shirt	22.0	555 units
GAR-004	Grey Hoodie	26.0	470 units
GAR-005	Women's Casual Top	14.0	875 units
GAR-006	Blue Denim Shirt	24.5	500 units
GAR-007	Green Sports T-Shirt	9.5	1,290 units
GAR-008	Beige Cargo Pants	28.0	437 units

The 85 percent utilisation rule

A line running at or above 85 percent sustained utilisation is treated as a bottleneck. This is not an arbitrary limit. Above 85 percent there is no slack to absorb absenteeism, machine breakdown or rework, so any disturbance propagates directly into the delivery date.

A bottleneck line must not be given additional volume without either a reallocation to another line or approved overtime. Planning a line above 95 percent is refused outright: the schedule will not survive contact with a single sick operator.

Efficiency ramp for new styles

A style running for the first time does not reach its target efficiency immediately. Plan the ramp as follows: day one at 60 percent, day two at 75 percent, day three at 85 percent, and full efficiency from day four.

For a short order of under 2,000 units the entire run may sit inside the ramp, so use 75 percent rather than 85 percent when quoting the date.

Overtime policy

Overtime is capped at two hours per operator per day and a maximum of three consecutive days. Beyond that, quality data shows the defect rate rising faster than the added output, which produces rework that costs more capacity than the overtime bought.

Overtime requires production manager approval and is never applied automatically by the planning system.